

WMO First Mile Data Management

World Meteorological Organization

Date: 2026-06-15

Version: 0.0.1-DRAFT-2026-06-15

Document location: TBD

Document status: DRAFT

Revision history: see [Annex E](#)

Standing Committee on Information Management and Technology (SC-IMT)^[1]

Commission for Observation, Infrastructure and Information Systems (INFCOM)^[2]

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i. Abstract

WMO regulates the real-time international exchange of observation data to ensure the availability of high-quality observational data for many applications in particular for meteorological and hydrological forecasts, and allowing Prediction Centers (for Numerical Weather Prediction – NWP as well as climate and hydrological prediction) to be able to produce reliable products. National Meteorological and Hydrological Services (NMHSs) use NWP products to monitor and predict the weather and hydrological conditions to issue forecasts and warnings for the protection of the lives and property of citizens against natural hazards, notably in the framework of the Early Warnings for All initiative.

NMHSs operate networks comprising different observing systems to meet the observational data requirements for national, regional and international needs, and to exchange data internationally, in accordance with the WMO technical regulations. Data are collected from the observing platforms, processed and quality controlled before being used nationally and exchanged internationally. WMO does not regulate this initial data collection segment of transmission of data from the observing station to the data collection and processing system (first-mile data collection).

This document defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

ii. Keywords

The following are keywords to be used by search engines and document catalogues.

wmo, observations, first mile, node, hub, protobuf, mqtt

iii. Security Considerations

TODO

No security considerations have been made for this standard.

Chapter 1. Scope

This document defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

This specification defines the conformance requirements for WMO First Mile Data Management (data model, encoding, transmission). Annex A defines the abstract test suite.

[1] <https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-int>

[2] <https://community.wmo.int/governance/commission-membership/infcom>

Chapter 2. Conformance

Conformance with this standard shall be checked using the tests specified in Annex A (normative) of this document.

WMO First Mile Data Management defines the encoding and transmission of observation data from remote monitoring stations (Nodes) to a central data collection system (Hub).

This standard identifies numerous Requirements Classes which define the functional requirements.

The mandatory Requirements Classes for this specification are:

- "Data Model"
- "Protocol"

Chapter 3. References

- Google LLC: Protocol Buffers Documentation (2026) ^[1]

[1] <https://protobuf.dev>

Chapter 4. Terms and definitions

This document uses the terms defined in [OGC Policy Directive 49](#), which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this Standard and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications ([OGC 08-131r3](#)), also known as the 'ModSpec'. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

The following additional terms and definitions also apply.

4.1. Abbreviated terms

Table 1. Symbols and abbreviated terms

Abbreviation	Term
CF	Climate and Forecast (CF) Metadata Conventions
HMEI	HydroMeteorological and Environmental Industry
MQP	Message Queuing Protocol
MQTT	Message Queuing Telemetry Transport
PubSub	Publish / Subscribe
QoS	Quality of Service
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
WIS	WMO Information System
WMO	World Meteorological Organization

Chapter 5. Conventions

This section provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of JSON Schema, or special notes regarding how to read the document.

5.1. Identifiers

The normative provisions in this Standard are denoted by the URI:

<http://wis.wmo.int/spec/fmdm/1>

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

5.2. Examples

Protocol topic examples provided in this specification are encoded as **plain text strings**.

Data and metadata message examples provided in this specification are encoded as JSON.

Complete examples can be found at <https://schemas.wmo.int/fmdm/1.0.0/examples>

5.3. Schemas

Monitoring event message schemas can be found at <https://schemas.wmo.int/fmdm/1.0.0>

5.4. Schema representation

Protobuf[<https://protobuf.dev>] constructs are used throughout this standard to define the structure of data and metadata messages.

Data and metadata message instances are always defined as JSON.

Chapter 6. Introduction

TODO

Chapter 7. Data Model

TODO one sentence

7.1. Requirements Class "Data Model"

7.1.1. Overview

This Requirements Class provides requirements for the Data Model.

Requirements Class	
http://wis.wmo.int/spec/fmdm/1/req/data-model	
Target type	Event metadata
Dependency	Protobuf
Pre-conditions	The event message conforms to the Protobuf specification.

Chapter 8. Protocol

TODO one sentence

8.1. Requirements Class "Protocol"

8.1.1. Overview

This Requirements Class provides requirements for the Protocol.

Unresolved directive in sections/clause_8_protocol.adoc -
include::../requirements/requirements_class_protocol.adoc[]

Annex A: Conformance Class Abstract Test Suite (Normative)

A.1. Conformance Class: Data Model

label

<http://wis.wmo.int/spec/fmdm/1/req/data-model>

subject

Requirements Class "Data Model"

classification

Target Type:Event Metadata

A.1.1. Encoding

label

/conf/data-model/encoding

subject

/req/data-model/encoding

test-purpose

Validate that a data model is encoding as Protobuf.

TODO

TODO

A.2. Conformance Class: Protocol

Unresolved directive in sections/annex_a_ats.adoc -
include::../abstract_tests/ATS_class_protocol.adoc[]

Annex B: Schemas (Normative)

NOTE

Schema documents will only be published on schemas.wmo.int once the standard has been approved.

B.1. Data Model

```
syntax = "proto3";

package wmo.firstmile.poc1;

option java_package = "wmo.firstmile.poc1";
option java_outer_classname = "Firstmile";

import "google/protobuf/empty.proto";
import "google/protobuf/timestamp.proto";

enum CellMethod {
    CELL_METHOD_UNSPECIFIED = 0; // Required default
    POINT = 1; // Instantaneous value at a point
    SUM = 2; // Sum over the cell
    MAXIMUM = 3; // Maximum value
    MAXIMUM_ABSOLUTE_VALUE = 4; // Maximum of absolute values
    MEDIAN = 5; // Median value
    MID_RANGE = 6; // Average of maximum and minimum
    MINIMUM = 7; // Minimum value
    MINIMUM_ABSOLUTE_VALUE = 8; // Minimum of absolute values
    MEAN = 9; // Mean (average) value
    MEAN_ABSOLUTE_VALUE = 10; // Mean of absolute values
    MEAN_OF_UPPER_DECILE = 11; // Mean of the upper group of data values defined by
the upper tenth of their distribution
    MODE = 12; // Mode (most common value)
    RANGE = 13; // Absolute difference between maximum and minimum
    ROOT_MEAN_SQUARE = 14; // Root mean square (RMS)
    STANDARD_DEVIATION = 15; // Standard deviation within the cell
    SUM_OF_SQUARES = 16; // Sum of squares
    VARIANCE = 17; // Variance within the cell
}

message DeviceRef {
    oneof target {
        google.protobuf.Empty node = 1; // if field is present => refers to the single
node
        uint32 observerId = 2;
    }
}

message Parameter {
```

```

string longName = 1;
string unit = 2;
map<string, string> standardNames = 3;
DeviceRef device = 4;
CellMethod cellMethod = 5;
uint32 cellPeriodSeconds = 6;
}

message ParameterDefinition {
    uint32 id = 1;
    string description = 2;
    repeated Parameter parameters = 3;
}

message Value {
    oneof kind {
        float floatValue = 1;
        double doubleValue = 2;
        int32 intValue = 3;
        uint32 unsignedIntValue = 4;
        int64 int64Value = 5;
        uint64 unsignedInt64Value = 6;
        string stringValue = 7;
        bool boolValue = 8;
        google.protobuf.Empty emptyValue = 9;
    }
}

message Observation {
    uint32 parameterDefinitionId = 1;
    google.protobuf.Timestamp time = 2;
    repeated Value values = 3;
}

enum ReferenceSurface {
    REFERENCE_SURFACE_UNSPECIFIED = 0; // Default
    MSL = 1;                          // Mean Sea Level
    GEOID = 2;                         // Geoid
    GL = 3;                           // Ground Level
    REFERENCE_ELLIPSOID = 4;          // Reference Ellipsoid
    PRESSURE_1000_HPA = 5;            // 1000 hPa Pressure Level
}

message Location {
    optional double latitude = 1;
    optional double longitude = 2;
    double heightMeter = 3;
    ReferenceSurface referenceSurface = 4;
}

message Node {

```

```

    string name = 1;
    optional Location location = 2;
    optional string url = 3;
    optional string serialNumber = 4;
    optional string firmwareVersion = 5;
}

message ObserverDevice {
    uint32 id = 1;
    string name = 2;
    optional Location location = 3;
    optional string url = 4;
    optional string serialNumber = 5;
    optional string firmwareVersion = 6;
}

message Data {
    repeated Observation observations = 1;
}

message Metadata {
    Node node = 1;
    repeated ObserverDevice observers = 2;
    repeated ParameterDefinition parameterDefinitions = 3;
    map<string, string> namespaces = 4;
}

message FirstMileMessage {
    oneof content {
        // MUST be published with retained flag
        Metadata metadata = 1;

        // MUST NOT be published with retained flag
        Data data = 2;
    }
}

```


Annex C: Examples (Informative)

C.1. Transmission topics

C.2. Data and metadata messages

Annex D: Bibliography

- IANA: Link Relation Types, <https://www.iana.org/assignments/link-relations/link-relations.xml>

Annex E: Revision History

Date	Release	Editor	Primary clauses modified	Description
2026-06-15	Template	Tom Kralidis	all	initial revision